Rectilinear Kinematics *(Ch. 12)*

$v = \dfrac{ds}{dt}, \quad a = \dfrac{dv}{dt}, \quad a\,ds = v\,dv$ last one when $a = f(s)$

Constant a_c (pick the one that leaves out what you neither know nor want):

$v = v_0 + a_c t$ no s

$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$ no v

$v^2 = v_0^2 + 2a_c(s - s_0)$ no t

$s = s_0 + \frac{1}{2}(v_0 + v)t$ no a_c

Free fall: $a_c = -g = -9.8\ \text{m/s}^2$ $+y$ up; $v = 0$ at max height

Graphs:

v = slope of s-t, $\quad a$ = slope of v-t

Δv = area under a-t, $\quad \Delta s$ = area under v-t

Variable acceleration:

$a = f(t)$: $\; v = v_0 + \displaystyle\int_0^t a\,dt, \quad s = s_0 + \int_0^t v\,dt$

$a = f(v)$: $\; t = \displaystyle\int_{v_0}^{v} \frac{dv}{f(v)}, \quad s - s_0 = \int_{v_0}^{v} \frac{v\,dv}{f(v)}$

$a = f(s)$: $\; v^2 = v_0^2 + 2\displaystyle\int_{s_0}^{s} a\,ds$

$v = f(s)$: $\; a = v\dfrac{dv}{ds}, \quad t = \displaystyle\int_{s_0}^{s} \frac{ds}{v}$

Direction reversal: solve $v = 0$ for the turning time and split the interval there. Total distance $\neq$ displacement.

Curvilinear and Projectile Motion *(Ch. 12)*

$\mathbf{v} = \dfrac{d\mathbf{r}}{dt}$ always tangent to the path

$\mathbf{a} = \dfrac{d\mathbf{v}}{dt}$ generally NOT tangent to the path

$v_x = \dot{x},\ v_y = \dot{y},\ v_z = \dot{z}; \quad a_x = \ddot{x},\ a_y = \ddot{y},\ a_z = \ddot{z}$ each axis independent

Path given as $y = f(x)$ (chain rule):

$v_y = \dfrac{dy}{dx}\, v_x$

$a_y = \dfrac{d^2y}{dx^2}\, v_x^2 + \dfrac{dy}{dx}\, a_x$

$\theta = \tan^{-1}\left(\dfrac{dy}{dx}\right)$ slope of the path, for n-t force components

Projectile ($a_x = 0$, $a_y = -g$), linked by common t:

$x = x_0 + (v_x)_0 t$

$y = y_0 + (v_y)_0 t - \frac{1}{2} g t^2$

$v_y = (v_y)_0 - gt, \quad v_y^2 = (v_y)_0^2 - 2g(y - y_0)$

Shortcuts, launch speed v_0 at θ_0 above horizontal:

$y = x\tan\theta_0 - \dfrac{gx^2}{2v_0^2\cos^2\theta_0}$ trajectory, no t

$R = \dfrac{v_0^2 \sin 2\theta_0}{g}, \quad t_f = \dfrac{2v_0 \sin\theta_0}{g}$ land at launch height only

$H = \dfrac{v_0^2 \sin^2\theta_0}{2g}$ max height ($v_y = 0$)

$v = \sqrt{v_x^2 + v_y^2}, \quad \theta = \tan^{-1}(v_y/v_x)$ from x-axis

Normal-Tangential (n-t) Coordinates *(Ch. 12)*

$\mathbf{v} = \dot{s}\,\hat{\mathbf{u}}_t, \qquad \mathbf{a} = a_t\hat{\mathbf{u}}_t + a_n\hat{\mathbf{u}}_n$

$a_t = \dot{v} = v\dfrac{dv}{ds}$ changes the speed

$a_n = \dfrac{v^2}{\rho}$ changes direction; points to centre of curvature

$a = \sqrt{a_t^2 + a_n^2}$

$\rho = \dfrac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ path $y = f(x)$

Circle: $\rho = r,\ s = r\theta,\ v = r\dot{\theta},\ a_t = r\ddot{\theta},\ a_n = r\dot{\theta}^2$ θ in rad

Straight line: $\rho \to \infty,\ a_n = 0$

Polar and Cylindrical Coordinates *(Ch. 12)*

$\mathbf{r} = r\hat{\mathbf{u}}_r$

$v_r = \dot{r}, \quad v_\theta = r\dot{\theta}, \quad v = \sqrt{v_r^2 + v_\theta^2}$

$a_r = \ddot{r} - r\dot{\theta}^2$

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ $2\dot{r}\dot{\theta}$ matters whenever r varies

$a = \sqrt{a_r^2 + a_\theta^2}$

$\dot{\hat{\mathbf{u}}}_r = \dot{\theta}\,\hat{\mathbf{u}}_\theta, \quad \dot{\hat{\mathbf{u}}}_\theta = -\dot{\theta}\,\hat{\mathbf{u}}_r$ unit-vector rates

Path given as $r = f(\theta)$:

$\dot{r} = \dfrac{dr}{d\theta}\dot{\theta}, \quad \ddot{r} = \dfrac{d^2r}{d\theta^2}\dot{\theta}^2 + \dfrac{dr}{d\theta}\ddot{\theta}$

$\tan\psi = \dfrac{r}{dr/d\theta}$ radial line to tangent; slotted arm, cam

$v_z = \dot{z}, \quad a_z = \ddot{z}$ cylindrical

$r = \text{const}$: $a_r = -r\dot{\theta}^2,\ a_\theta = r\ddot{\theta}$

Dependent and Relative Motion *(Ch. 12)*

$s_A + l_{CD} + s_B = l_T$ position coordinates from fixed datums

$v_A + v_B = 0, \quad a_A + a_B = 0$ differentiate; same coefficients

Coefficient = number of cord segments that change length at that moving pulley. Opposite signs: blocks move oppositely.

$\dfrac{d}{dt}\sqrt{x^2 + h^2} = \dfrac{x\dot{x}}{\sqrt{x^2 + h^2}}$ cord over a pulley at height h

$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$ translating axes on A

$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}, \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$ solve in components

If B is on a curve, $\mathbf{a}_B$ has both $(a_B)_t$ and $(a_B)_n = v_B^2/\rho$.

$\dfrac{A}{\sin a} = \dfrac{B}{\sin b} = \dfrac{C}{\sin c}$ sine law

$C^2 = A^2 + B^2 - 2AB\cos c$ cosine law

Equations of Motion *(Ch. 13)*

$\Sigma\mathbf{F} = m\mathbf{a}$ FBD and kinetic diagram first

$W = mg; \quad$ FPS: $m = W/g$ slug $= \text{lb}\cdot\text{s}^2/\text{ft}$ $g = 32.2\ \text{ft/s}^2$

$F_k = \mu_k N$ opposes sliding

$F_s \le \mu_s N$ equal only at impending slip

$F_{sp} = ks$ s from unstretched length

$\Sigma\mathbf{F}_{ext} = m_{tot}\,\mathbf{a}_G$ system of particles

Rectangular (inclines, drag, thrust):

$\Sigma F_x = ma_x, \quad \Sigma F_y = ma_y, \quad \Sigma F_z = ma_z$

Normal-tangential (known ρ: curves, loops, pendulums):

$\Sigma F_t = m\dfrac{dv}{dt}, \quad \Sigma F_n = m\dfrac{v^2}{\rho}, \quad \Sigma F_b = 0$

Pendulum or loop recipe:

1) energy from the known point gives v^2; 2) $\Sigma F_n = mv^2/\rho$ gives T or N.

$T - mg\cos\theta = m\dfrac{v^2}{l}$ cord length l, θ from the lowest point

Cord slack: $T = 0$; $\quad$ leaves surface: $N = 0$

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Top of vertical loop: $v_{min} = \sqrt{g\rho}$

Polar and cylindrical (rotating slotted arm):

$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2), \quad \Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$

$\Sigma F_z = m\ddot{z}$

$\dfrac{d^2u}{d\theta^2} + u = -\dfrac{F}{mh^2u^2}, \quad u = \dfrac{1}{r}$ $F > 0$ away from O

$h = r^2\dot{\theta} = rv_\theta = \text{const}$ any central force

Gravitation and Orbits *(Ch. 13 to 15)*

$F = \dfrac{GMm}{r^2}$ toward the centre

$G = 66.73(10^{-12})\ \text{m}^3/(\text{kg}\cdot\text{s}^2), \quad M_e = 5.976(10^{24})\ \text{kg}$

$R_e = 6378\ \text{km}$ use the value the question gives

$GM_e = gR_e^2 \approx 3.988(10^{14})\ \text{m}^3/\text{s}^2$

$r = R_e + h$ questions give altitude h, not r

Circular orbit ($\Sigma F_n = mv^2/r$):

$v = \sqrt{\dfrac{GM}{r}}, \quad T = \dfrac{2\pi r}{v} = 2\pi\sqrt{\dfrac{r^3}{GM}}$

$r = \left(\dfrac{GMT^2}{4\pi^2}\right)^{1/3}$ geosynchronous: $T = 23.934\ \text{h} = 86\,162\ \text{s}$

Energy and momentum (any orbit):

$V_g = -\dfrac{GMm}{r}$ far from the surface

$\frac{1}{2}v_A^2 - \dfrac{GM}{r_A} = \frac{1}{2}v_B^2 - \dfrac{GM}{r_B}$

$r_Av_A = r_Bv_B$ perigee A, apogee B ($\mathbf{v} \perp \mathbf{r}$)

$v_A = \sqrt{\dfrac{2GM\,r_B}{r_A(r_A + r_B)}}$ speed at A from r_A, r_B

$v_e = \sqrt{\dfrac{2GM}{r}}$ escape speed

Work and Energy *(Ch. 14)*

$U_{1-2} = \displaystyle\int_{s_1}^{s_2} F\cos\theta\, ds$ + if F has a component along motion

$U_W = -W\Delta y$ Δy positive upward

$U_s = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right)$ by spring on body; − when deformation grows

Work done ON the spring $= -U_s$. Stretch and compression both count as deformation s.

$U_f = -\mu_k N\,\Delta s$ friction, constant N

$T = \frac{1}{2}mv^2$

$\Sigma T_1 + \Sigma U_{1-2} = \Sigma T_2$ work-energy

Connected blocks: apply to the whole system with the cord constraint; tension in an inextensible cord does no net work.

$P = \dfrac{dU}{dt} = \mathbf{F}\cdot\mathbf{v}, \quad P = Fv$ watts (J/s)

$\varepsilon = \dfrac{P_{out}}{P_{in}} < 1, \quad P_{in} = \dfrac{P_{out}}{\varepsilon}$ efficiency

$V_g = Wy$ + above datum, − below

$V_e = \frac{1}{2}ks^2 \geq 0$

$T_1 + V_1 = T_2 + V_2$ only if all working forces are conservative

Impulse and Momentum *(Ch. 15)*

$m\mathbf{v}_1 + \Sigma\displaystyle\int_{t_1}^{t_2}\mathbf{F}\,dt = m\mathbf{v}_2$

$m(v_x)_1 + \Sigma\displaystyle\int F_x\,dt = m(v_x)_2$ same for y

$\displaystyle\int F\,dt$ = area under F-t graph

Starting from rest with friction: nothing moves until $F = \mu_sN$. Find that t_1, then integrate from t_1 with μ_kN.

$t = \dfrac{v}{\mu_k g}$ slide to rest on level ground

$F_{avg} = \dfrac{m(v_2 - v_1)}{\Delta t}$

$\Sigma m(v)_1 = \Sigma m(v)_2$ no external impulse

During impact, weight and friction are non-impulsive: neglect them.

$v = \dfrac{m_Av_A + m_Bv_B}{m_A + m_B}$ plastic impact, coupling ($e = 0$)

$e = \dfrac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$ $e = 1$ elastic, $e = 0$ plastic

$\Delta T = \Sigma T_1 - \Sigma T_2$ energy lost in impact

Oblique impact: momentum and e along the line of impact n; each v_t unchanged (smooth).

$\mathbf{H}_O = \mathbf{r}\times m\mathbf{v}, \quad H_O = mvd$ d = perpendicular distance

$H_O = mr^2\dot{\theta} = mrv_\theta$

$\mathbf{H}_O = \Sigma\,\mathbf{r}_i \times m_i\mathbf{v}_i$ system

$$\mathbf{r}\times m\mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ r_x & r_y & r_z \\ mv_x & mv_y & mv_z \end{vmatrix}$$

$(H_O)_1 + \Sigma\displaystyle\int_{t_1}^{t_2} M_O\,dt = (H_O)_2$ same fixed O

$M_z = F_t\,r, \quad H_z = mv_t\,r$ only F_t has a moment about z

H_O is conserved when $\Sigma M_O = 0$ (central force, orbits).

Units

$1\ \text{ft} = 0.3048\ \text{m}, \quad 1\ \text{mi} = 5280\ \text{ft}$

$60\ \text{mi/h} = 88\ \text{ft/s}, \quad \text{km/h} \div 3.6 = \text{m/s}$

$1\ \text{Mm/h} = 277.8\ \text{m/s}, \quad 1\ \text{Mg} = 1000\ \text{kg}$

$1\ \text{lb} = 4.448\ \text{N}, \quad 1\ \text{slug} = 14.59\ \text{kg}$

$1\ \text{hp} = 550\ \text{ft}\cdot\text{lb/s} = 745.7\ \text{W}$

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